circuit. The intensity of the waves at a point is proportional to the fourth power of the frequency. The situation with respect to the matter waves of beta radiation is quite analogous, with the sole difference that their energy is proportional not to the fourth power of the frequency, but—as shown by a closer theoretical analysis by Fermi—to its sixth power. It follows, therefore, that the decay probability (the energy emitted per second divided by the energy of the individual decay) is proportional to the fifth power of the energy available for the decay process.

The above considerations represent the actual conditions in their general outlines only, but not quantitatively correctly. Before we can gain a complete understanding of the subject, we must discuss still another consideration. We have assumed the dipole moment of the aerial to be constant. However, this cannot be expected to be the case for every one of the various nuclear types. On the contrary, considerable individual differences must be expected to exist, and do in fact exist. One would therefore expect the decay probability to be the product of two factors, the first being determined by the dipole moment (and as the latter is determined quantitatively by the size of the nucleus in question, it may vary widely from one nucleus to another), and the second one being proportional to the fifth power of the decay energy. Figure 23 shows a summary of the lifetimes, and hence of the decay probabilities and energies, of the beta emitters. Whenever a power of a quantity is supposed to be proportional to a power of another quantity, it is most convenient to record on the co-ordinate axes not these powers themselves but their logarithms, for the logarithms must be linearly related to each other. Thus, Figure 23 shows *log T*, the logarithm of the half-life, *T*, measured in seconds, as a function of *log E*; the scales on the two axes differ by the factor 5. If the fifth-power law is valid, all the beta emitters must lie on a straight line forming an angle of 135 deg. with the axes. Actually, the empirical points lie, mostly, between two such straight lines; the distance between these two straight lines provides a measure of the fluctuation of the dipole moment from one nucleus to another.

Before proceeding to the discussion of other decay processes, let us write down the formulae of a few more processes involving both alpha and beta radiation. When the uranium atom ${}_{92}U^{238}$ decays under emission of an alpha particle, ${}_{2}He^{4}$, the product of the process is an atom of mass number 234 and atomic number 90—uranium X_1, ${}_{90}UX_1{}^{234}$. We write this process as follows:

$$ {}_{92}U^{238} \rightarrow {}_{90}UX_1{}^{234} + {}_{2}He^{4} $$